

Discrete gradient-zeroing neural dynamics for future Moore–Penrose inverse with application to tracking control of manipulator

Wenqi Wu, Yunong Zhang*

School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China

Key Laboratory of Machine Intelligence and Advanced Computing, Ministry of Education, Guangzhou 510006, China

ARTICLE INFO

Keywords:

Gradient-zeroing neural dynamics (GZND)
Zhang time discretization (ZTD) formula
Extrapolation formula
Future Moore–Penrose inverse
Tracking control

ABSTRACT

To handle the future Moore–Penrose inverse problem effectively, the continuous gradient-zeroing neural dynamics (CGZND) model without inverse computing is proposed via combining the advantages of the zeroing neural dynamics (ZND) model and the gradient neural dynamics (GND) model. In order to readily implement in computer and digital hardware, the order-6 discrete gradient-zeroing neural dynamics (O6-DGZND) algorithm is further proposed via the simultaneous combination of the order-6 Zhang time discretization (O6-ZTD) formula and the order-6 extrapolation (O6-E) formula. The O6-DGZND algorithm is not only highly accurate, but also utilized to solve the challenging Z-type future Moore–Penrose inverse problem. Besides, four other DGZND algorithms with low-precision are also presented to meet different precision requirements in actual engineering applications. The convergence properties of the DGZND algorithms, including the 0-stability, consistency, and truncation error, are theoretically analyzed. Numerical experiment results illustrate the efficacy and validity of the proposed DGZND algorithms, and the superiority of the O6-DGZND algorithm is also verified. In addition, with the aid of the O6-DGZND algorithm, an application on a UR10 robot manipulator tracking task is performed successfully, which implies that the practicability of the O6-DGZND algorithm in real-world applications.

1. Introduction

The Moore–Penrose inverse (MPI, also called pseudoinverse), as a generalization of matrix inverse, plays a critical role in mathematics and engineering. According to whether the matrix $M(t) \in \mathbb{C}^{m \times n}$ changes with time, the MPI problem is classified as time-independent (also called static) MPI problem and time-dependent MPI problem. In particular, the time-dependent matrix $M(t) \in \mathbb{C}^{m \times n}$ reduces to constant matrix $M \in \mathbb{C}^{m \times n}$ when the matrix is not changed with time, that is, the static MPI problem is a special case of the time-dependent MPI problem. Therefore, the investigation for the time-dependent MPI problem is more representative and significant than that for the static MPI problem. The definition and related properties of time-dependent MPI are presented as follows.

Definition 1 (Rao and Mitra, 1971). For a given time-dependent matrix $M(t) \in \mathbb{C}^{m \times n}$, then there exists a unique matrix $X(t) \in \mathbb{C}^{n \times m}$ satisfying the following four Penrose equations:

$$\begin{aligned} M(t)X(t)M(t) &= M(t), & X(t)M(t)X(t) &= X(t), \\ (M(t)X(t))^H &= M(t)X(t), & (X(t)M(t))^H &= X(t)M(t), \end{aligned}$$

where H is the conjugate transpose operator. The matrix $X(t) \in \mathbb{C}^{n \times m}$ to be obtained is called the time-dependent MPI of matrix $M(t)$, and denoted by $M^\dagger(t)$.

Specifically, when the matrix $M(t) \in \mathbb{C}^{m \times n}$ is full-rank, its MPI $M^\dagger(t)$ can be obtained via the following lemma.

Lemma 1 (Rao and Mitra, 1971; Wang, 1997). For a matrix $M(t) \in \mathbb{C}^{m \times n}$ being of full rank, the MPI $M^\dagger(t)$ is explicitly written as

$$M^\dagger(t) = \begin{cases} (M^H(t)M(t))^{-1}M^H(t), & \text{if } m > n, \\ M^{-1}(t), & \text{if } m = n, \\ M^H(t)(M(t)M^H(t))^{-1}, & \text{if } m < n. \end{cases}$$

With the rapid development of industrial technology, the application of MPI is increasingly widespread, such as manipulator control (Jin et al., 2019; Liao & Liu, 2015), signal processing (Van der Veen et al., 1997), pattern recognition (Zhang, Wu et al., 2021), and image recovery (Chountasis et al., 2009). In light of its critical role, much significance has been attached to obtaining the online solution of the MPI problem. For the time-independent MPI computing, a group of

* Corresponding author at: School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China.

E-mail addresses: wuwq53@mail2.sysu.edu.cn (W. Wu), zhynong@mail.sysu.edu.cn (Y. Zhang).

numerical methods with serial computation nature have been developed, discussed, and studied (Hartwig, 1976; Petkovic & Stanimirovic, 2011; Zhou et al., 2002), such as Newton iterations method (Petkovic & Stanimirovic, 2011), SVD method (Hartwig, 1976), and Greville's recursive method (Zhou et al., 2002). Although these methods are performed well in time-independent MPI problem solving, their limitations are exposed in large-scale or time-dependent MPI problem solving because of the serial nature (Wang, 1997; Wu & Zheng, 2020). Therefore, the neural dynamics method with parallel computation characteristic has become a preferred tool for handling the MPI problem. In the last several years, recurrent neural dynamics has gained much development in solving various scientific computing and applications (Jin, Wei & Li, 2023; Jin et al., 2020; Qin et al., 2017; Zhang, Li et al., 2018; Zhang et al., 2019a). As a kind of recurrent neural dynamics, the gradient neural dynamics (GND, also termed gradient neural network), which is generated in the negative gradient direction, was proposed to compute the time-independent problems (Zhang et al., 2009). However, for the time-dependent problems computing, the essence of GND leads to the data unable to be updated in time, and consequently generates large lagging error. Hence, the GND model is ineffective even fails to handle the time-dependent problems (Liu et al., 2022).

In order to overcome the limitation, differing from the design of the GND, another class of recurrent neural dynamics called zeroing neural dynamics (ZND, also termed Zhang neural dynamics) was proposed by Zhang et al. (2002), which has been applied to handling various time-dependent zero finding problems effectively (Chen, Li & Lin, 2020; Jia et al., 2020; Jin et al., 2017; Jin, Zheng & Luo, 2022; Kong et al., 2023; Li & Li, 2014; Sun et al., 2022; Wu & Zheng, 2022). Moreover, the ZND model possesses excellent convergence performance, e.g., global and exponential convergence (Li & Zhang, 2010). For further improving convergence rate and robustness of ZND model, many variants of ZND model for solving time-dependent problem were developed, investigated, and enhanced (Jin et al., 2019; Liu et al., 2023; Wu & Zhang, 2022; Xiao et al., 2021, 2022; Zhang, Kong et al., 2018; Zhang & Zheng, 2019). On the basis of the ZND design method, five continuous ZND models for computing continuous time-dependent MPI problem were provided by Liao and Zhang (2014). In particular, one of which called ZND-V model is the only explicit dynamics model, and the ZND-V model is often referred to as the GMDS model because it can be also generated from the Getz-Marsden dynamics system (GMDS) (Getz & Marsden, 1997).

With the development of modern science, solving objects become more and more complex, particularly in time-discrete and real-time cases, as well as cases with unknown future information. The continuous ZND method is far from being satisfactory for solving these systems. In view of the knotty problem, previous studies have successively proposed a group of Zhang time discretization (ZTD) formulas with different precision (Chen, Guo & Zhang, 2020; Chen & Zhang, 2019; Euler, 1755; Guo et al., 2017; Jin & Zhang, 2014), e.g., order-2 ZTD (O2-ZTD), order-3 ZTD (O3-ZTD), order-4 ZTD (O4-ZTD), order-5 ZTD (O5-ZTD), and order-6 ZTD (O6-ZTD) formulas, to discretize the continuous ZND model. The resultant discrete ZND (DZND) algorithms are directly and readily implemented in computer and digital hardware. The DZND algorithms for handling various future discrete problems have been developed, analyzed, and investigated (Jin & Zhang, 2014; Li et al., 2020; Qiu & Zhang, 2019). It is worth pointing out that, the O2-ZTD formula, i.e., the classical Euler-forward formula (Euler, 1755), is regarded as the simplest and the least accurate ZTD formula (Qiu & Zhang, 2019). In this paper, the discrete time-dependent MPI problem, especially the future MPI problem is investigated. For the future MPI problem, the continuous matrix $M(t)$ in Definition 1 is replaced with the discrete $M_k = M(t_k)$, in which t_k denotes the k th time sampling instant, and the next or future coefficient matrix M_{k+1} is unknown. Following Zhang, Shi et al. (2018), the future MPI problem is classified as the Getz-Marsden-type future MPI problem and the Zhang-type future MPI problem according to the solvability by the discrete

GMDS (DGMDS) algorithm. In other words, the future MPI problem of which exact MPI can be solved by the DGMDS algorithm is termed GM-type future MPI problem, the other one of which exact MPI cannot be obtained by the DGMDS algorithm is called Z-type future MPI problem. Unfortunately, most previous algorithms are only applied to cope with GM-type future MPI problems, it is still a challenging task to solve Z-type future MPI problems by those algorithms. Hence, the achievement for the challenging task is one of the objectives we considered in this paper.

With the aid of ZTD formulas, some DZND algorithms for the future MPI problem were proposed, analyzed, and studied (Guo et al., 2017; Li et al., 2019; Qiu & Zhang, 2019; Wu et al., 2020; Zhang, Li et al., 2021; Zhang et al., 2022). For example, in 2017, Guo et al. presented the DGMDS algorithm with precision of $O(\tau^3)$ to solve the future matrix inversion (Guo et al., 2017), where $O(\tau^3)$ denotes the pattern matrix with each element being order $O(\tau^3)$ and τ represents sampling gap. In 2019, for solving the future matrix inversion, the DGMDS algorithm with precision of $O(\tau^4)$ was further developed (Qiu & Zhang, 2019). The high-precision GMDS-ZNN algorithm with $O(\tau^6)$ was also proposed (Wu et al., 2020). However, those three algorithms are designed for the future matrix inversion solving but not the generalized future MPI. In order to compute the future MPI problem including the Z-type future MPI, the algorithm with order of $O(\tau^3)$ was discussed (Zhang et al., 2022). To further improve the computational accuracy, the other algorithm with the precision of $O(\tau^5)$ was presented (Zhang, Li et al., 2021).

In this paper, a novel continuous model for computing the continuous time-dependent MPI, i.e., continuous gradient-zeroing neural dynamics (CGZND) model, is firstly proposed by combining the GND and ZND models. Moreover, on the basis of the simultaneous O6-ZTD and order-6 extrapolation (O6-E) formulas, the resultant discrete gradient-zeroing neural dynamics (DGZND) algorithm, i.e., order-6 DGZND (O6-DGZND) algorithm, which possesses higher precision $O(\tau^6)$ than the aforementioned algorithms, is proposed, analyzed, and investigated. It is worth pointing out that the O6-DGZND algorithm is not only high accurate, but also utilized to handle the more challenging Z-type future MPI problem effectively. For comparison, with the aid of the combination of ZTD and extrapolation formulas, four other DGZND algorithms, i.e., the O5-DGZND, O4-DGZND, O3-DGZND, and O2-DGZND algorithms, are also provided, analyzed, and investigated so as to meet different precision requirements in actual engineering applications.

At the end of this section, the main contributions of this paper are listed as below.

- (1) For handling the future MPI problem effectively, we firstly propose a CGZND model without inverse computing. Sequentially, to further adapt in the digital computers and improve the computational accuracy, five DGZND algorithms termed as O2-DGZND through O6-DGZND are proposed by using different ZTD and extrapolation formulas.
- (2) The good convergence performance of the DGZND algorithms is analyzed and demonstrated via rigorous mathematical derivations. It also deduces that the O6-DGZND algorithm has higher precision due to the ingenious combination of O6-ZTD and O6-E formulas.
- (3) Comparative experiments are illustrated to substantiate the validity and accuracy of the proposed DGZND algorithms. The superiority of O6-DGZND algorithm is also substantiated. Remarkably, the proposed DGZND algorithms can also be utilized to solve the knotty Z-type future MPI problem while the existing DGMDS algorithm cannot.
- (4) The simulation experiment on the basis of a UR10 robot manipulator is performed to verify the effectiveness and feasibility of the O6-DGZND algorithm in real-world applications.

Table 1
Main abbreviations.

MPI	Moore–Penrose inverse
ZND	Zeroing neural dynamics
GND	Gradient neural dynamics
CGZND	Gradient-zeroing neural dynamics
CGZND	Continuous gradient-zeroing neural dynamics
DGZND	Discrete gradient-zeroing neural dynamics
GMDS	Getz-Marsden dynamics system
DGMDS	Discrete Getz-Marsden dynamics system
O6-, O5-, O4-, O3-, O2- DGZND	Order-6, -5, -4, -3, -2 discrete gradient-zeroing neural dynamics
O6-, O5-, O4-, O3-, O2- ZTD	Order-6, -5, -4, -3, -2 Zhang time discretization
O6-, O5-, O4-, O3-, O2- E	Order-6, -5, -4, -3, -2 extrapolation

To improve the readability of this paper, the research route and main abbreviations of this paper are presented in Fig. 1 and Table 1, respectively. Moreover, the organization of this paper is as follows. The problem formulation and CGZND model are provided in Section 2. In Section 3, five DGZND algorithms are proposed and studied. Meanwhile, the convergence performance of the DGZND algorithms is proved by rigorous mathematical deduction. Illustrative and comparative experiments shown in Section 4 substantiate the efficacy and superiority of the O6-DGZND algorithm. In Section 5, the application to a UR10 robot manipulator is carried out to validate the practicability of the O6-DGZND algorithm. Finally, Section 6 concludes this paper.

2. Problem formulation and CGZND model

We focus on computing the future MPI of the full rank matrix at each computational time interval $[t_k, t_{k+1}) \subset [0, t_f] \subset [0, +\infty)$. Because of the similarity and space limitation, only the future MPI of full row rank is considered in this paper.

2.1. Problem formulation

The future MPI X_{k+1} of full row rank to be computed is described as

$$X_{k+1} = M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1} \in \mathbb{C}^{n \times m} \quad (1)$$

during the computational time interval $[t_k, t_{k+1}) \subset [0, t_f] \subset [0, +\infty)$, where t_f represents the task duration; τ denotes the sampling gap; $k \in \mathbb{Z}^+$ represents the updating index; $t_k = k\tau$ is the k th time sampling instant. Full-row-rank matrix $M_{k+1} \in \mathbb{C}^{m \times n}$ is unknown, and it is generated from the previous data or information (e.g., M_k and M_k^H). The goal of this paper is to obtain the future MPI X_{k+1} of matrix M_{k+1} .

2.2. CGZND model

To compute future MPI (1) effectively, the corresponding continuous time-dependent MPI problem needs to be presented and investigated firstly. According to Lemma 1, the unknown MPI $X(t)$ of time-dependent matrix $M(t)$ is described as

$$X(t) = M^H(t)(M(t)M^H(t))^{-1} \in \mathbb{C}^{n \times m}, \quad (2)$$

where $M(t) \in \mathbb{C}^{m \times n}$ is full-row-rank and smooth at $[0, t_f]$.

Differing from previous regular work (Liao & Zhang, 2014) with the knotty inverse computing problem, firstly, one focuses on obtaining the inverse $(M(t)M^H(t))^{-1}$ via designing a new model without inverse computing, i.e., CGZND model. Let $Z(t) = (M(t)M^H(t))^{-1} \in \mathbb{C}^{m \times m}$. According to the ZND method, the error function is chosen as $E(t) = M(t)M^H(t) - Z^{-1}(t)$ to compute $(M(t)M^H(t))^{-1}$. Applying ZND design formula $\dot{E}(t) = -\lambda E(t)$ with $\lambda \in \mathbb{R}^+$, one has

$$-\dot{Z}^{-1}(t) = -\dot{M}(t)M^H(t) - M(t)\dot{M}^H(t) - \lambda(M(t)M^H(t) - Z^{-1}(t)), \quad (3)$$

where $\dot{Z}^{-1}(t) = d(Z^{-1}(t))/dt$ and parameter $\lambda \in \mathbb{R}^+$. In the physical sense, parameter λ generally corresponds to the reciprocal of the capacitance parameter (Zhang & Yi, 2011). Hence, parameter λ should

be set as large as the hardware would permit, or appropriately large for practical application purposes. Evidently, (3) is an implicit dynamic equation, and the following lemma is presented to obtain the explicit dynamic equation.

Lemma 2 (Zhang et al., 2022). Consider a time-dependent nonsingular matrix $B(t)$, the time derivative of its inverse matrix $B^{-1}(t)$ denoted by $\dot{B}^{-1}(t)$ can be obtained via the following equation:

$$\dot{B}^{-1}(t) = -B^{-1}(t)\dot{B}(t)B^{-1}(t).$$

With the help of Lemma 2, (3) is transformed into

$$\begin{aligned} \dot{Z}(t) = & -Z(t)(\dot{M}(t)M^H(t) + M(t)\dot{M}^H(t))Z(t) \\ & - \lambda(Z(t)M(t)M^H(t) - I_{n \times n})Z(t), \end{aligned} \quad (4)$$

in which $I_{n \times n} \in \mathbb{R}^{n \times n}$ is an $n \times n$ identity matrix.

Secondly, let $\beta(t) = \|Z(t)M(t)M^H(t) - I_{n \times n}\|_F^2/2$. Applying GND design formula, i.e., $\dot{Z}(t) = -\gamma\partial\beta(t)/\partial Z(t)$, yields the following GND model:

$$\dot{Z}(t) = -\gamma(Z(t)M(t)M^H(t) - I_{n \times n})M(t)M^H(t), \quad (5)$$

where $\gamma \in \mathbb{R}^+$ and $\|\cdot\|_F$ denotes Frobenius norm of a matrix.

According to the observation for the last term on the right of (4) (i.e., $-\lambda(Z(t)M(t)M^H(t) - I_{n \times n})Z(t)$) and the term on the right of (5) (i.e., $-\gamma(Z(t)M(t)M^H(t) - I_{n \times n})M(t)M^H(t)$), one obtains that the difference between them is that the latter contains more known information, which may be more effective and accurate in the solution process. Therefore, replacing the last term on the right of (4) with the term on the right of (5), the following model is presented:

$$\begin{aligned} \dot{Z}(t) = & -Z(t)(\dot{M}(t)M^H(t) + M(t)\dot{M}^H(t))Z(t) \\ & - \gamma(Z(t)M(t)M^H(t) - I_{n \times n})M(t)M^H(t). \end{aligned} \quad (6)$$

Finally, the CGZND model is further described as

$$\begin{cases} Z(0) = Z_0, \\ \dot{Z}(t) = -Z(t)(\dot{M}(t)M^H(t) + M(t)\dot{M}^H(t))Z(t) \\ \quad - \gamma(Z(t)M(t)M^H(t) - I_{n \times n})M(t)M^H(t), \\ X(t) = M^H(t)Z(t), \end{cases} \quad (7)$$

where $Z_0 \in \mathbb{C}^{m \times m}$ is the initial state.

Theorem 1. Suppose that time-dependent matrix $M(t)$ is full-row-rank for any $t \in [0, t_f]$. Starting from an initial state Z_0 , the state matrix $X(t)$ synthesized by CGZND model (7) globally and exponentially converges to the theoretical time-dependent MPI of matrix $M(t)$.

Proof. The proof process is omitted due to the similarity with the proof of Lemma 3 in the paper of Zhang, Li et al. (2021). $\square$

3. DGZND algorithms

In order to make CGZND model (7) more readily reproduced on the computer to solve future MPI problem (1), in this section, with the aid of the O6-ZTD and O6-E formulas, the high-precision O6-DGZND algorithm is proposed. Moreover, four other DGZND algorithms are given in the same manner.

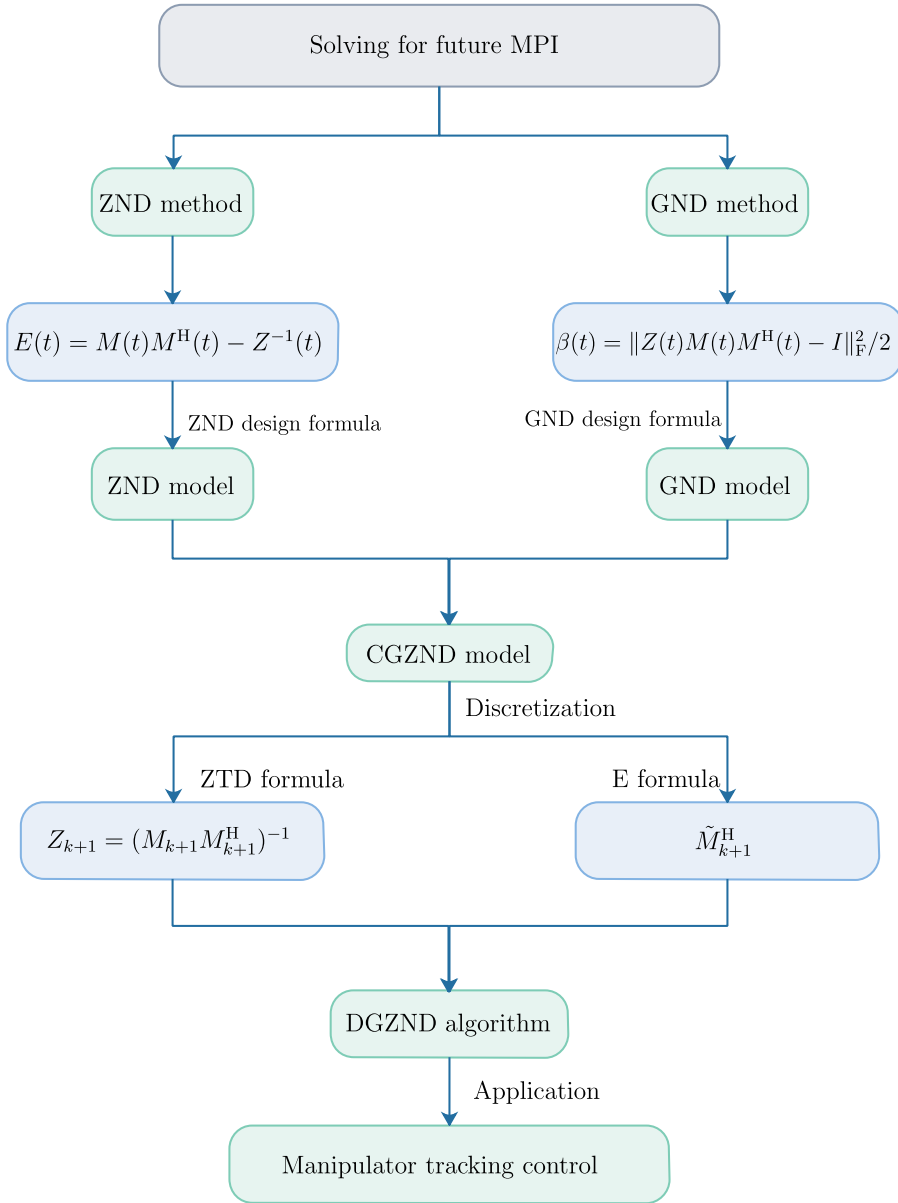


Fig. 1. Paper research route.

3.1. O6-DGZND algorithm

In this subsection, the O6-ZTD and O6-E discretization formulas are firstly presented, and the corresponding O6-DGZND algorithm is proposed.

Firstly, on the basis of the following O6-ZTD formula, the future Z_{k+1} can be obtained by utilizing the present and previous data information, i.e., $Z_k, Z_{k-1}, \dots, Z_{k-8}$. The O6-ZTD formula (Chen, Guo & Zhang, 2020) is formulated as

$$\dot{y}(t_k) = \frac{1}{\tau} \sum_{i=0}^9 b_i y(t_{k+1-i}) + O(\tau^5), \quad (8)$$

where $O(\tau^5)$ represents the remainder and updating index $k = 8, 9, 10, \dots$. In this paper, the coefficients b_i with $i = 0, 1, \dots, 9$ are set as the first row of Table 1 in Chen, Guo and Zhang (2020), i.e., $b_0 = 350/841$, $b_1 = 11471/235480$, $b_2 = -140/841$, $b_3 = -280/841$, $b_4 = -350/2523$, $b_5 = 70/841$, $b_6 = 784/4205$, $b_7 = -70/2523$, $b_8 = -680/5887$, and $b_9 = 315/6728$.

Besides, O6-ZTD formula (8) is rewritten as

$$y(t_{k+1}) = \frac{\tau}{b_0} \dot{y}_k - \frac{1}{b_0} \sum_{i=1}^9 b_i y(t_{k+1-i}) + O(\tau^6). \quad (9)$$

Therefore, continuous model (6) is discretized into the following algorithm by using the O6-ZTD formula (9):

$$\begin{aligned} Z_{k+1} \doteq & -\frac{h}{b_0} (Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H - \frac{\tau}{b_0} Z_k (\dot{M}_k M_k^H \\ & + M_k \dot{M}_k^H) Z_k - \frac{1}{b_0} \sum_{i=1}^9 b_i Z_{k+1-i}, \end{aligned} \quad (10)$$

where $\doteq$ is the computational assignment operator, and $h = \tau\gamma$ denotes the step length.

Secondly, to obtain the future M_{k+1}^H , the O6-E formula is presented in the following lemma.

Lemma 3 (Zhang et al., 2019b). Suppose that the function $s(t)$ is smooth, and $\{s(t_i)|t_i = i\tau, i = 0, 1, 2, \dots\}$ represents its time instant sequence with $\tau \in (0, 1)$ s denoting sampling gap. Then, the following equation holds true:

Algorithm 1: O6-DGZND Algorithm (12) for Future MPI Solving

```

1. Initialize:  $Z_0$  and  $M_0$ ;
2. Set:  $t_f$ ,  $\tau$  and  $h$ ;
3. for  $k = 0 : (t_f/\tau_k - 1)$  do
4.   if  $k = 0 : 8$  then
5.      $Z_{k+1} = -h(Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H - \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k + Z_k$ ;
6.      $\tilde{M}_{k+1}^H = 2M_k^H - M_{k-1}^H$ ;
7.      $X_{k+1} = \tilde{M}_{k+1}^H Z_{k+1}$ ;
8.   else
9.      $Z_{k+1} = -\frac{h}{b_0} (Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H - \frac{\tau}{b_0} Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k$ 
        $-\frac{1}{b_0} \sum_{i=1}^9 b_i Z_{k+1-i}$ ;
10.     $\tilde{M}_{k+1}^H = 6M_k^H - 15M_{k-1}^H + 20M_{k-2}^H - 15M_{k-3}^H + 6M_{k-4}^H - M_{k-5}^H$ ;
11.     $X_{k+1} = \tilde{M}_{k+1}^H Z_{k+1}$ ;
12.  end if
13.  Update:  $k := k + 1$ ;
14.  Return;
15. end for

```

$$s(t_{k+1}) = 6s(t_k) - 15s(t_{k-1}) + 20s(t_{k-2}) - 15s(t_{k-3}) + 6s(t_{k-4}) - s(t_{k-5}) + O(\tau^6), \quad (11)$$

where $k \in \mathbb{Z}$ and $k \geq 5$.

Proof. The proof is presented in Appendix A. $\square$

On the basis of Lemma 3, the future M_{k+1}^H is approximated by the known M_{k-j}^H , $j = 0, 1, \dots, 5$ as follows:

$$\tilde{M}_{k+1}^H \doteq 6M_k^H - 15M_{k-1}^H + 20M_{k-2}^H - 15M_{k-3}^H + 6M_{k-4}^H - M_{k-5}^H.$$

Finally, combining O6-ZTD formula (9) and O6-E formula (11), one proposes the O6-DGZND algorithm as

$$\begin{cases} Z_0 = Z(0), \\ Z_{k+1} \doteq -\frac{h}{b_0} (Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H - \frac{\tau}{b_0} Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k \\ \quad + M_k \dot{M}_k^H Z_k - \frac{1}{b_0} \sum_{i=1}^9 b_i Z_{k+1-i}, \\ \tilde{M}_{k+1}^H \doteq 6M_k^H - 15M_{k-1}^H + 20M_{k-2}^H - 15M_{k-3}^H + 6M_{k-4}^H - M_{k-5}^H, \\ X_{k+1} \doteq \tilde{M}_{k+1}^H Z_{k+1}, \end{cases} \quad (12)$$

in which, matrix $\tilde{M}_{k+1}^H$ is the approximation of matrix M_{k+1}^H . In addition, the pseudo code about O6-DGZND algorithm (12) is presented in Algorithm 1.

The convergence performance of O6-DGZND algorithm (12), including 0-stability, consistency, truncation error, and maximal steady-state residual error (MSSRE), is analyzed in the following theorem. Note that the residual error is defined as $e_{k+1} = \|X_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F$, and the MSSRE is denoted as $\lim_{k \rightarrow +\infty} \sup e_{k+1}$.

Theorem 2. With the sufficiently small sampling gap $\tau \in (0, 1)$ s, let $O(\tau^6)$ denote the pattern matrix with each element being $O(\tau^6)$. Suppose that the matrix $M_k \in \mathbb{C}^{m \times n}$ is full-row-rank for any $t_k \in [0, t_f]$. Then, O6-DGZND algorithm (12) is 0-stable, consistent, and convergent, which converges with the order of truncation error being $O(\tau^6)$. Besides, O6-DGZND algorithm (12) converges toward the theoretical MPI of M_k with the MSSRE $\lim_{k \rightarrow +\infty} \sup e_{k+1}$ being of order $O(\tau^6)$.

Proof. Let z_{k+1}^{ij} and $\tilde{m}_{k+1}^{ij}$ denote the ij th element of matrices $Z_{k+1} \in \mathbb{C}^{m \times m}$ and $\tilde{M}_{k+1}^H \in \mathbb{C}^{n \times m}$, respectively. Following O6-DGZND algorithm (12), the Z_{k+1} is formulated as

$$Z_{k+1} = \frac{\tau}{b_0} \dot{Z}_k - \frac{1}{b_0} \sum_{s=1}^9 b_s Z_{k+1-s}.$$

Therefore, the matrix X_{k+1} is expressed as

$$X_{k+1} = \tilde{M}_{k+1}^H Z_{k+1} = \tilde{M}_{k+1}^H \left(\frac{\tau}{b_0} \dot{Z}_k - \frac{1}{b_0} \sum_{s=1}^9 b_s Z_{k+1-s} \right),$$

and its ij th element is written as

$$\sum_{l=1}^m \tilde{m}_{k+1}^{il} z_{k+1}^{lj} = \sum_{l=1}^m \tilde{m}_{k+1}^{il} \left[\left(\frac{\tau}{b_0} z_k^{lj} - \frac{1}{b_0} \sum_{s=1}^9 b_s z_{k+1-s}^{lj} \right) \right]. \quad (13)$$

Rearranging (13) yields

$$u_{k+1} + \sum_{s=1}^9 c_s u_{k+1-s} = \frac{\tau}{b_0} \dot{u}_k,$$

where $u_{k+1} = \sum_{l=1}^m \tilde{m}_{k+1}^{il} z_{k+1}^{lj}$, $\dot{u}_k = \sum_{l=1}^m \tilde{m}_{k+1}^{il} \dot{z}_k^{lj}$, $u_{k+1-s} = \sum_{l=1}^m \tilde{m}_{k+1}^{il} z_{k+1-s}^{lj}$, and $c_s = b_s/b_0$. According to Appendix B, the characteristic polynomial of O6-DGZND algorithm (12) is the same as that of O6-ZTD formula (9), and is described as

$$\Gamma_9(t) = t^9 + \sum_{s=1}^9 c_s t^{9-s}.$$

Following Chen, Guo and Zhang (2020), one knows that the chosen values d_i with $i = 0, 1, \dots, 9$ such that all the roots of characteristic equation $\Gamma_9(t) = 0$ satisfy the root condition of Result 1 in Appendix B. Hence, according to Result 1 in Appendix B, O6-DGZND algorithm (12) is 0-stable.

Let $\bar{M}_{k+1}^H$ and $\bar{Z}_{k+1}$ denote the exact value of $\tilde{M}_{k+1}^H$ and Z_{k+1} , respectively. According to O6-ZTD formula (9) and O6-E formula (11), one has

$$\begin{aligned} M_{k+1}^\dagger &= \bar{M}_{k+1}^H \bar{Z}_{k+1} \\ &= (6M_k^H - 15M_{k-1}^H + 20M_{k-2}^H - 15M_{k-3}^H + O(\tau^6)) \\ &\quad \times \left(\frac{\tau}{b_0} \dot{Z}_k - \frac{1}{b_0} \sum_{s=1}^9 b_s Z_{k+1-s} + O(\tau^6) \right) \\ &= \bar{M}_{k+1}^H Z_{k+1} + O(\tau^6) = X_{k+1} + O(\tau^6), \end{aligned} \quad (14)$$

which implies the truncation error of O6-DGZND algorithm (12) is $O(\tau^6)$. Therefore, according to Result 2 in Appendix B, O6-DGZND algorithm (12) is consistent of order-5. Finally, on the basis of Result 3 in Appendix B, it is concluded that O6-DGZND algorithm (12) is convergent, which converges with the order of truncation error being $O(\tau^6)$.

In addition, the MSSRE $\lim_{k \rightarrow +\infty} \sup e_{k+1}$ synthesized by O6-DGZND algorithm (12) is derived as follows:

$$\begin{aligned} & \lim_{k \rightarrow +\infty} \sup \|X_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F \\ &= \lim_{k \rightarrow +\infty} \sup \|(\bar{M}_{k+1}^H + O(\tau^6))(\bar{Z}_{k+1} + O(\tau^6)) - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F \\ &\leq \lim_{k \rightarrow +\infty} \sup \|\bar{M}_{k+1}^H \bar{Z}_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F \\ &\quad + \lim_{k \rightarrow +\infty} \sup \|\bar{M}_{k+1}^H O(\tau^6) + O(\tau^6) \bar{Z}_{k+1} + O(\tau^{12})\|_F \\ &\leq \lim_{k \rightarrow +\infty} \sup (\|\bar{M}_{k+1}^H\|_F + \|\bar{Z}_{k+1}\|_F) O(\tau^6) = O(\tau^6). \end{aligned}$$

That is, the MSSRE $\lim_{k \rightarrow +\infty} \sup e_{k+1}$ synthesized by O6-DGZND algorithm (12) is $O(\tau^6)$. The proof is therefore completed. $\square$

3.2. O5-DGZND algorithm

In this subsection, the O5-ZTD formula and O5-E formula are presented, and the corresponding O5-DGZND algorithm is proposed.

Firstly, the O5-ZTD formula (Chen & Zhang, 2019) is presented as

$$\begin{aligned} y(t_{k+1}) &= \frac{51}{1000} y(t_k) + \frac{2}{5} y(t_{k-1}) + \frac{3}{5} y(t_{k-2}) + \frac{1}{5} y(t_{k-3}) - \frac{7}{40} y(t_{k-4}) \\ &\quad - \frac{22}{125} y(t_{k-5}) + \frac{1}{10} y(t_{k-6}) + \frac{111}{50} \tau \dot{y}(t_k) + O(\tau^5), \end{aligned} \quad (15)$$

where the updating index $k = 6, 7, 8, \dots$. Therefore, continuous model (6) is further discretized into

$$\begin{aligned} Z_{k+1} &\doteq -\frac{111}{50} h(Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H \\ &\quad - \frac{111}{50} \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k + \frac{51}{1000} Z_k \\ &\quad + \frac{2}{5} Z_{k-1} + \frac{3}{5} Z_{k-2} + \frac{1}{5} Z_{k-3} - \frac{7}{40} Z_{k-4} - \frac{22}{125} Z_{k-5} + \frac{1}{10} Z_{k-6}, \end{aligned} \quad (16)$$

where h is defined the same as before.

Secondly, the O5-E formula is presented by the following lemma.

Lemma 4 (Zhang et al., 2019b). Suppose that the function $s(t)$ is smooth, and $\{s(t_i)|t_i = i\tau, i = 0, 1, 2, \dots\}$ represents its time instant sequence with $\tau \in (0, 1)$ s denoting sampling gap. Then, the following equation holds true:

$$s(t_{k+1}) = 5s(t_k) - 10s(t_{k-1}) + 10s(t_{k-2}) - 5s(t_{k-3}) + s(t_{k-4}) + O(\tau^5), \quad (17)$$

where $k \in \mathbb{Z}$ and $k \geq 4$.

Proof. The proof is omitted due to the similarity with that of Lemma 3. $\square$

According to Lemma 4, on the basis of the present and previous data M_k^H, M_{k-1}^H , and M_{k-3}^H , the future M_{k+1}^H can be updated as

$$\tilde{M}_{k+1}^H \doteq 5M_k^H - 10M_{k-1}^H + 10M_{k-2}^H - 5M_{k-3}^H + M_{k-4}^H.$$

Finally, the O5-DGZND algorithm is formulated as

$$\begin{cases} Z_0 = Z(0), \\ Z_{k+1} \doteq -\frac{111}{50} h(Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H \\ \quad - \frac{111}{50} \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k + \frac{51}{1000} Z_k \\ \quad + \frac{2}{5} Z_{k-1} + \frac{3}{5} Z_{k-2} + \frac{1}{5} Z_{k-3} - \frac{7}{40} Z_{k-4} - \frac{22}{125} Z_{k-5} + \frac{1}{10} Z_{k-6}, \\ \tilde{M}_{k+1}^H \doteq 5M_k^H - 10M_{k-1}^H + 10M_{k-2}^H - 5M_{k-3}^H + M_{k-4}^H, \\ X_{k+1} \doteq \tilde{M}_{k+1}^H Z_{k+1}. \end{cases} \quad (18)$$

The convergence performance of O5-DGZND algorithm (18), including 0-stability, consistency, and truncation error, is analyzed in the following theorem.

Theorem 3. With the sufficiently small sampling gap $\tau \in (0, 1)$ s , let $O(\tau^5)$ denote the pattern matrix with each element being $O(\tau^5)$. Suppose that the matrix $M_k \in \mathbb{C}^{m \times n}$ is full-row-rank for any $t_k \in [0, t_f]$. Then, O5-DGZND algorithm (18) is 0-stable, consistent, and convergent, which converges with the order of truncation error being $O(\tau^5)$. Besides, O5-DGZND algorithm (18) converges toward the theoretical MPI of M_k with the MSSRE being of order $O(\tau^5)$.

Proof. The proof is omitted due to the similarity with that of Theorem 2. $\square$

3.3. O4-DGZND algorithm

In this subsection, the O4-ZTD and O4-E discretization formulas are presented to discretize continuous model (6), and the discrete O4-DGZND algorithm is proposed.

Firstly, the O4-ZTD formula (Guo et al., 2017) is formulated as

$$\begin{aligned} y(t_{k+1}) &= \frac{5}{24} y(t_k) + \frac{1}{2} y(t_{k-1}) + \frac{1}{4} y(t_{k-2}) + \frac{1}{6} y(t_{k-3}) \\ &\quad - \frac{1}{8} y(t_{k-4}) - 2\tau \dot{y}(t_k) + O(\tau^4), \end{aligned} \quad (19)$$

where the updating index $k = 4, 5, 6, \dots$. Then, continuous model (6) is discretized into the following algorithm by utilizing O4-ZTD formula (19):

$$\begin{aligned} Z_{k+1} &\doteq -2h(Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H - 2\tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k \\ &\quad + \frac{5}{24} Z_k + \frac{1}{2} Z_{k-1} + \frac{1}{4} Z_{k-2} + \frac{1}{6} Z_{k-3} - \frac{1}{8} Z_{k-4}, \end{aligned}$$

where h is defined the same as before.

Secondly, the O4-E formula is presented by the following lemma.

Lemma 5 (Zhang et al., 2019b). Suppose that the function $s(t)$ is smooth, and $\{s(t_i)|t_i = i\tau, i = 0, 1, 2, \dots\}$ represents its time instant sequence with $\tau \in (0, 1)$ s denoting sampling gap. Then, the following equation holds true:

$$s(t_{k+1}) = 4s(t_k) - 6s(t_{k-1}) + 4s(t_{k-2}) - s(t_{k-3}) + O(\tau^4), \quad (20)$$

where $k \in \mathbb{Z}$ and $k \geq 3$.

Proof. The proof is omitted due to the similarity with that of Lemma 3. $\square$

According to Lemma 5, the future M_{k+1}^H is approximated by the known M_k^H, M_{k-1}^H , and M_{k-3}^H as follows:

$$\tilde{M}_{k+1}^H \doteq 4M_k^H - 6M_{k-1}^H + 4M_{k-2}^H - M_{k-3}^H.$$

Finally, the O4-DGZND algorithm is presented as

$$\begin{cases} Z_0 = Z(0), \\ Z_{k+1} \doteq -2h(Z_k M_k M_k^H - I_{n \times n}) M_k M_k^H - 2\tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H) Z_k \\ \quad + \frac{5}{24} Z_k + \frac{1}{2} Z_{k-1} + \frac{1}{4} Z_{k-2} + \frac{1}{6} Z_{k-3} - \frac{1}{8} Z_{k-4}, \\ \tilde{M}_{k+1}^H \doteq 4M_k^H - 6M_{k-1}^H + 4M_{k-2}^H - M_{k-3}^H, \\ X_{k+1} \doteq \tilde{M}_{k+1}^H Z_{k+1}. \end{cases} \quad (21)$$

The convergence performance of O4-DGZND algorithm (21) is analyzed in the following theorem.

Theorem 4. With the sufficiently small sampling gap $\tau \in (0, 1)$ s , let $O(\tau^4)$ denote the pattern matrix with each element being $O(\tau^4)$. Suppose that the matrix $M_k \in \mathbb{C}^{m \times n}$ is full-row-rank for any $t_k \in [0, t_f]$. Then, O4-DGZND algorithm (21) is 0-stable, consistent, and convergent, which converges with the order of truncation error being $O(\tau^4)$. Besides, O4-DGZND algorithm (21) converges toward the theoretical MPI of M_k with the MSSRE being of order $O(\tau^4)$.

Proof. The proof is omitted due to the similarity with that of [Theorem 2](#). $\square$

3.4. O3-DGZND algorithm

Firstly, the O3-ZTD formula, also termed Taylor-Zhang ZTD formula ([Jin & Zhang, 2014](#)), is formulated as

$$y(t_{k+1}) = \frac{3}{2}y(t_k) - y(t_{k-1}) + \frac{1}{2}y(t_{k-2}) + \tau \dot{y}(t_k) + O(\tau^3), \quad (22)$$

where the updating index $k = 2, 3, 4, \dots$. On the basis of O3-ZTD formula (22), continuous model (6) is discretized into the following algorithm.

$$\begin{aligned} Z_{k+1} \doteq & -h(Z_k M_k M_k^H - I_{n \times n})M_k M_k^H - \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H)Z_k \\ & + \frac{3}{2}Z_k - Z_{k-1} + \frac{1}{2}Z_{k-2}, \end{aligned} \quad (23)$$

where h is defined the same as before.

To obtain the future matrix M_{k+1}^H , the O3-E formula is proposed in the following lemma.

Lemma 6 ([Zhang et al., 2019b](#)). Suppose that the function $s(t)$ is smooth, and $\{s(t_i)|t_i = i\tau, i = 0, 1, 2, \dots\}$ represents its time instant sequence with $\tau \in (0, 1)$ denoting sampling gap. Then, the following equation holds true:

$$s(t_{k+1}) = 3s(t_k) - 3s(t_{k-1}) + s(t_{k-2}) + O(\tau^3), \quad (24)$$

where $k \in \mathbb{Z}$ and $k \geq 2$.

Proof. The proof is omitted due to the similarity with that of [Lemma 3](#). $\square$

With the aid of [Lemma 6](#), the future matrix M_{k+1}^H is approximated as

$$\tilde{M}_{k+1}^H \doteq 3M_k^H - 3M_{k-1}^H + M_{k-2}^H.$$

Finally, the O3-DGZND algorithm is described as

$$\begin{cases} Z_0 = Z(0), \\ Z_{k+1} \doteq -h(Z_k M_k M_k^H - I_{n \times n})M_k M_k^H - \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H)Z_k \\ \quad + \frac{3}{2}Z_k - Z_{k-1} + \frac{1}{2}Z_{k-2}, \\ \tilde{M}_{k+1}^H \doteq 3M_k^H - 3M_{k-1}^H + M_{k-2}^H, \\ X_{k+1} \doteq \tilde{M}_{k+1}^H Z_{k+1}. \end{cases} \quad (25)$$

Additionally, the convergence performance of O3-DGZND algorithm (25) is analyzed in the following theorem.

Theorem 5. With the sufficiently small sampling gap $\tau \in (0, 1)$ s, let $O(\tau^3)$ denote the pattern matrix with each element being $O(\tau^3)$. Suppose that the matrix $M_k \in \mathbb{C}^{m \times n}$ is full-row-rank for any $t_k \in [0, t_f]$. Then, O3-DGZND algorithm (25) is 0-stable, consistent, and convergent, which converges with the order of truncation error being $O(\tau^3)$. Besides, O3-DGZND algorithm (25) converges toward the theoretical MPI of M_k with the MSSRE being of order $O(\tau^3)$.

Proof. The proof is omitted due to the similarity with that of [Theorem 2](#). $\square$

3.5. O2-DGZND algorithm

In this subsection, on the basis of the previous information, i.e., Z_k and Z_{k-1} , the future Z_{k+1} is approximated by the O2-ZTD formula, and the O2-ZTD formula ([Euler, 1755](#)) is described as

$$y(t_{k+1}) = y(t_k) + \tau \dot{y}(t_k) + O(\tau^2), \quad (26)$$

where the updating index $k = 0, 1, 2, \dots$. Thus, the continuous model (6) is discretized into

$$\begin{aligned} Z_{k+1} \doteq & -h(Z_k M_k M_k^H - I_{n \times n})M_k M_k^H \\ & - \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H)Z_k + Z_k, \end{aligned} \quad (27)$$

where h is defined the same as before.

Subsequently, the O2-E formula is presented in the following lemma:

Lemma 7 ([Zhang et al., 2019b](#)). Suppose that the function $s(t)$ is smooth, and $\{s(t_i)|t_i = i\tau, i = 0, 1, 2, \dots\}$ denotes its time instant sequence with $\tau \in (0, 1)$ denoting sampling gap. Then, the following equation holds true:

$$s(t_{k+1}) = 2s(t_k) - s(t_{k-1}) + O(\tau^2), \quad (28)$$

where $k \in \mathbb{Z}$ and $k \geq 1$.

Proof. The proof is omitted due to the similarity with that of [Lemma 3](#). $\square$

According to [Lemma 7](#), the future matrix M_k^H is approximated as

$$\tilde{M}_{k+1}^H \doteq 2M_k^H - M_{k-1}^H.$$

Finally, on the basis of O2-ZTD formula (26) and O2-E formula (28), the O2-DGZND algorithm is obtained as

$$\begin{cases} Z_0 = Z(0), \\ Z_{k+1} \doteq -h(Z_k M_k M_k^H - I_{n \times n})M_k M_k^H \\ \quad - \tau Z_k (\dot{M}_k M_k^H + M_k \dot{M}_k^H)Z_k + Z_k, \\ \tilde{M}_{k+1}^H \doteq 2M_k^H - M_{k-1}^H, \\ X_{k+1} \doteq \tilde{M}_{k+1}^H Z_{k+1}. \end{cases} \quad (29)$$

The convergence performance of O2-DGZND algorithm (29) is analyzed in the following theorem.

Theorem 6. With the sufficiently small sampling gap $\tau \in (0, 1)$ s, let $O(\tau^2)$ denote the pattern matrix with each element being $O(\tau^2)$. Suppose that the matrix $M_k \in \mathbb{C}^{m \times n}$ is full-row-rank for any $t_k \in [0, t_f]$. Then, O2-DGZND algorithm (29) is 0-stable, consistent, and convergent, which converges with the order of truncation error being $O(\tau^2)$. Besides, O2-DGZND algorithm (29) converges toward the theoretical MPI of M_k with the MSSRE being of order $O(\tau^2)$.

Proof. The proof is omitted due to the similarity with that of [Theorem 2](#). $\square$

Why combine the ON-ZTD formula and the ON-E formula with $N = 2, 3, 4, 5$, and 6? Taking O6-DGZND algorithm (12) as an example, the following two remarks can explain it.

Remark 1. Take O6-DGZND algorithm (12) as an example. When we combine O6-ZTD formula (9) and the aforementioned low-precision extrapolation formula, i.e., O5-E formula (17), O4-E formula (20), O3-E formula (24), and O2-E formula (28), to discretize CGZND model (7), similar to the proof of [Theorem 2](#), one concludes that the resultant DGZND algorithms are still 0-stable, consistent, and convergent, but converge with the order of truncation error being $O(\tau^5)$, $O(\tau^4)$, $O(\tau^3)$, and $O(\tau^2)$, respectively. Besides, the corresponding MSSREs synthesized by this algorithms are only achieve $O(\tau^5)$, $O(\tau^4)$, $O(\tau^3)$, and $O(\tau^2)$, respectively. In short, the resultant DGZND algorithm is of low precision once the precision of the utilized extrapolation formula is slightly lower than that of the ZTD formula.

Remark 2. On the other hand, when the aforementioned low-precision extrapolation formula is replaced with the high-precision extrapolation formula unmentioned in this paper, e.g., order-7 extrapolation formula, order-8 extrapolation formula, and even the higher-order extrapolation formula, to discretize CGZND model (7), similar to the proof of

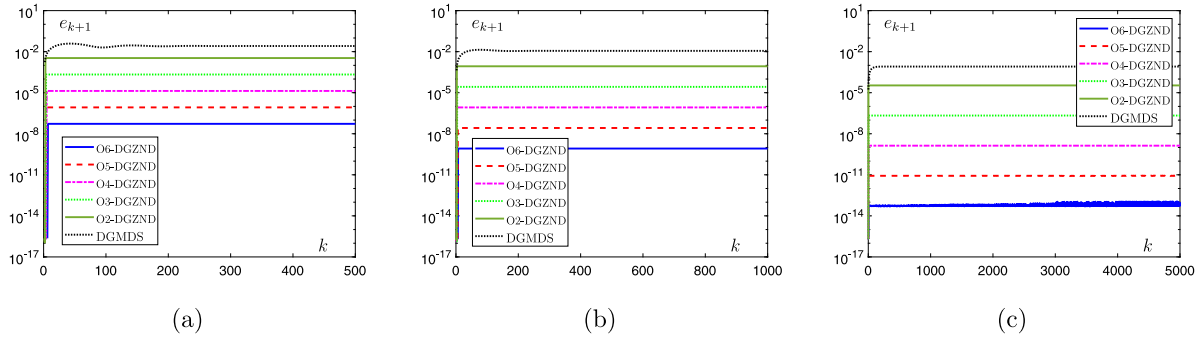


Fig. 2. Trajectories of residual errors $e_{k+1} = \|X_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F$ synthesized by DGMS and five DGZND algorithms with different τ for computing future MPI problem in Example 1. (a) With $\tau = 0.01$ s. (b) With $\tau = 0.005$ s. (c) With $\tau = 0.001$ s.

Table 2

Comparisons among Six Discrete Algorithms under Same Condition with Different τ in Terms of MSSRE, Convergence Precision (CP), and Consistency Order (CO) for Computing GM-type Future MPI Problem in Example 1.

Algorithm	Sampling Gap (s)	MSSRE	CP	CO	Convergence
DGMS (31)	$\tau = 0.01$	2.591×10^{-2}			Yes
	$\tau = 0.005$	1.148×10^{-2}	$\mathcal{O}(\tau^2)$	$\mathcal{O}(\tau)$	
	$\tau = 0.001$	8.154×10^{-4}			
O2-DGZND (29)	$\tau = 0.01$	3.418×10^{-3}			Yes
	$\tau = 0.005$	8.547×10^{-4}	$\mathcal{O}(\tau^2)$	$\mathcal{O}(\tau)$	
	$\tau = 0.001$	3.419×10^{-5}			
O3-DGZND (25)	$\tau = 0.01$	2.147×10^{-4}			Yes
	$\tau = 0.005$	2.685×10^{-5}	$\mathcal{O}(\tau^3)$	$\mathcal{O}(\tau^2)$	
	$\tau = 0.001$	2.148×10^{-7}			
O4-DGZND (21)	$\tau = 0.01$	1.349×10^{-5}			Yes
	$\tau = 0.005$	8.434×10^{-7}	$\mathcal{O}(\tau^4)$	$\mathcal{O}(\tau^3)$	
	$\tau = 0.001$	1.350×10^{-9}			
O5-DGZND (18)	$\tau = 0.01$	8.474×10^{-7}			Yes
	$\tau = 0.005$	2.650×10^{-8}	$\mathcal{O}(\tau^5)$	$\mathcal{O}(\tau^4)$	
	$\tau = 0.001$	8.515×10^{-12}			
O6-DGZND (12)	$\tau = 0.01$	5.323×10^{-8}			Yes
	$\tau = 0.005$	8.324×10^{-10}	$\mathcal{O}(\tau^6)$	$\mathcal{O}(\tau^5)$	
	$\tau = 0.001$	5.486×10^{-14}			

Theorem 2. one obtains that the resultant DGZND algorithm achieves 0-stability, consistency, and convergency. Besides, its truncation error and the MSSRE are still $\mathcal{O}(\tau^6)$ and $\mathcal{O}(\tau^6)$, respectively. However, the computational and storage costs are both increased because more previous data points are involved in. These results imply that the extrapolation formula of which precision is higher than that of the ZTD formula to discretize CGZND model (7) is in vain.

4. Numerical experiments and verification

In this section, two numerical experiments are conducted to illustrate the efficacy, high accuracy, and the superiority of O6-DGZND algorithm (12). In addition, the effectiveness and the precision pattern of four other DGZND algorithms are also verified.

For comparison, the representative GMS model for the continuous time-dependent MPI problem (2) is recalled (Liao & Zhang, 2014). According to the paper Liao and Zhang (2014) and ZND method, the GMS model is described as

$$\dot{X}(t) = -\lambda(X(t)M(t)X(t) - X(t)) - X(t)\dot{M}(t)X(t). \quad (30)$$

For computing the future MPI problem (1), continuous GMS model (30) is also discretized by O2-ZTD formula (26). The resultant DGMS algorithm is presented as

$$X_{k+1} = -h(X_k M_k X_k - X_k) - \tau X_k \dot{M}_k X_k + X_k, \quad (31)$$

which is utilized to compare with O6-DGZND algorithm (12) in the ensuing numerical experiments.

Example 1. Consider a specific GM-type future MPI with $M(t_k) \in \mathbb{C}^{2 \times 3}$ and

$$M(t_k) = \begin{bmatrix} \exp(i2\pi t_k) & \exp(i2\pi t_k) & -\exp(i2\pi t_k) \\ -\exp(i2\pi t_k) & \exp(i2\pi t_k) & \exp(i2\pi t_k) \end{bmatrix}.$$

The future MPI X_{k+1} of matrix $M_{k+1} = M(t_{k+1})$ is to be computed at each computational time interval $[k\tau, (k+1)\tau) \subset [0, t_f]$.

To verify the validity and superiority of O6-DGZND algorithm (12), DGMS algorithm and five proposed DGZND algorithms are applied to tackling this example. The task duration t_f is set as 5 s, the step length h is set as 0.02, as well as the sampling gap τ is chosen as 0.01 s, 0.005 s, and 0.001 s, respectively. The corresponding numerical experiment results are displayed in Fig. 2 and Table 2.

The trajectories of residual errors $e_{k+1} = \|X_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F$ synthesized by the DGMS and five DGZND algorithms are shown in Fig. 2(a) through (c) with different sampling gap τ , respectively. As displayed in Fig. 2, with the same condition, although the residual errors synthesized by six algorithms all converge toward 0, the magnitudes are different. Therefore, the effectiveness and validity of the six DGZND algorithms are well verified, and these algorithms have different accuracy. Specifically, the residual error synthesized by the proposed O6-DGZND algorithm (12) is the smallest, by contrast, DGMS algorithm (31) is the biggest, which implies that O6-DGZND algorithm (12) is the best in accuracy among six algorithms, and the GMS algorithm (31) is the worst.

Moreover, in order to better validate the precision patterns of six discrete algorithms, more details for MSSREs synthesized by six discrete algorithms with respect to different sampling gap values are shown in Table 2. As illustrated in Table 2, the MSSRE synthesized by O6-DGZND

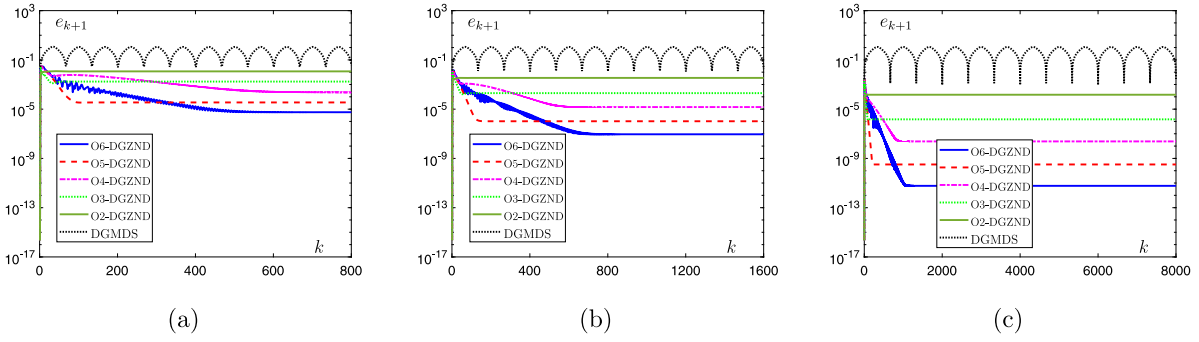


Fig. 3. Trajectories of residual errors $e_{k+1} = \|X_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F$ synthesized by DGMS and five DGZND algorithms with different τ for computing future MPI problem in Example 2. (a) With $\tau = 0.01$ s. (b) With $\tau = 0.005$ s. (c) With $\tau = 0.001$ s.

Table 3

Comparisons among Six Discrete Algorithms under Same Condition with Different Sampling Gap Values in Terms of MSSRE, Convergence Precision (CP), and Consistency Order (CO) for Computing Z-type Future MPI Problem in Example 2.

Algorithm	Sampling Gap (s)	MSSRE	CP	CO	Convergence
DGMS (31)	$\tau = 0.01$	1.111×10^0	—	—	No
	$\tau = 0.005$	1.123×10^0			
	$\tau = 0.001$	1.120×10^0			
O2-DGZND (29)	$\tau = 0.01$	1.168×10^{-2}	$O(\tau^2)$	$O(\tau)$	Yes
	$\tau = 0.005$	3.444×10^{-3}			
	$\tau = 0.001$	1.480×10^{-4}			
O3-DGZND (25)	$\tau = 0.01$	1.729×10^{-3}	$O(\tau^3)$	$O(\tau^2)$	Yes
	$\tau = 0.005$	1.995×10^{-4}			
	$\tau = 0.001$	1.485×10^{-6}			
O4-DGZND (21)	$\tau = 0.01$	2.228×10^{-4}	$O(\tau^4)$	$O(\tau^3)$	Yes
	$\tau = 0.005$	1.476×10^{-5}			
	$\tau = 0.001$	2.400×10^{-8}			
O5-DGZND (18)	$\tau = 0.01$	3.505×10^{-5}	$O(\tau^5)$	$O(\tau^4)$	Yes
	$\tau = 0.005$	1.040×10^{-6}			
	$\tau = 0.001$	3.223×10^{-10}			
O6-DGZND (12)	$\tau = 0.01$	5.638×10^{-6}	$O(\tau^6)$	$O(\tau^5)$	Yes
	$\tau = 0.005$	9.083×10^{-8}			
	$\tau = 0.001$	5.918×10^{-12}			

algorithm (12) with the sampling gap $\tau = 0.01$ s is 5.323×10^{-8} , the resultant MSSRE roughly reduced by a factor of 2^6 with τ decreasing by a factor of 2, i.e., τ from 0.01 s to 0.005 s. The MSSRE synthesized by O6-DGZND algorithm (12) reduces to the 5.486×10^{-14} when τ decreases from 0.01 s to 0.001 s. Besides, this regular pattern is also followed when τ reduces from 0.005 s to 0.001 s. Hence, the MSSRE synthesized by O6-DGZND algorithm (12) changes in the pattern of $O(\tau^6)$, and O6-DGZND algorithm (12) converges to the theoretical MPI with the MSSRE being of order $O(\tau^6)$. In addition, one can similarly obtain that the MSSREs synthesized by O5-DGZND algorithm (18), O4-DGZND algorithm (21), O3-DGZND algorithm (25), and O2-DGZND algorithm (29) are of the orders of $O(\tau^5)$, $O(\tau^4)$, $O(\tau^3)$, and $O(\tau^2)$, respectively. Besides, the DGMS algorithm (31) is of $O(\tau^2)$. In summary, the validity of five DGZND algorithms and the correctness of theoretical results are well illustrated, as well as the superiority of O6-DGZND algorithm (12) compared with other algorithms.

Example 2. Consider a specific Z-type future MPI with $M(t_k) \in \mathbb{C}^{2 \times 3}$ and

$$M(t_k) = \begin{bmatrix} \exp(i2\pi t_k) & \exp(i2\pi t_k) & -\exp(i2\pi t_k) \\ -\exp(i2\pi t_k) & \exp(i2\pi t_k) & \exp(i5\pi t_k) \end{bmatrix}.$$

The future MPI X_{k+1} of $M_{k+1} = M(t_{k+1})$ is to be computed at each computational time interval $[k\tau, (k+1)\tau) \subset [0, t_f]$ s.

With the task duration $t_f = 8$ s, the step length $h = 0.02$, and the sampling gap τ set as 0.01 s, 0.005 s, and 0.001 s, respectively, the DGMS algorithm and five DGZND algorithms are adopted to solve this example. The corresponding results are displayed in Figs. 3 and 4 as well as Table 3.

Specifically, the trajectories of the residual errors $e_{k+1} = \|X_{k+1} - M_{k+1}^H (M_{k+1} M_{k+1}^H)^{-1}\|_F$ synthesized by the DGMS and five DGZND algorithms are shown in Fig. 3. According to Fig. 3, one observes that the residual errors e_{k+1} synthesized by five DGZND algorithms are all relatively accurately converge to 0, especially for O6-DGZND algorithm (12), which can extremely near to 0 with a relative small sampling gap τ . By contrast, the e_{k+1} synthesized by DGMS algorithm (31) is far from 0 and even more than 1, even though when the sampling gap τ is reduced and the updating index is increased. To show the ineffectiveness of DGMS algorithm (31) better in the Z-type future MPI problem solving, the elemental trajectories $x_{k+1,ij} = a_{ij} + ib_{ij}$ are illustrated in Fig. 4 with $\tau = 0.01$ s, where a_{ij} and b_{ij} respectively denote its real part and imaginary part of the element $x_{k+1,ij}$ synthesized by DGMS algorithm (31). As seen from this figure, a visible distance between element $x_{k+1,ij}$ and its theoretical value $x_{k+1,ij}^*$ is generated, especially for the $x_{k+1,22}$. Hence, one concludes that five DGZND algorithms can be utilized to compute the Z-type future MPI problem effectively, in which O6-DGZND algorithm (12) has excellent performance in terms of the calculation accuracy under the same condition. By comparison, DGMS algorithm (31) fails to handle the Z-type future MPI problem, not to mention the calculation accuracy.

To better verify the correctness of theoretical results, more detailed MSSREs data synthesized by five DGZND algorithms are displayed in Table 3 with respect to different τ . As shown in Table 3, similar to Example 1, one also concludes that the MSSRE synthesized by the proposed O6-DGZND algorithm (12) changes in an $O(\tau^6)$ pattern. Besides, the ones by the O5-DGZND (18), O4-DGZND (21), O3-DGZND (25), and O2-DGZND (29) are changing in $O(\tau^5)$, $O(\tau^4)$, $O(\tau^3)$, and $O(\tau^2)$,

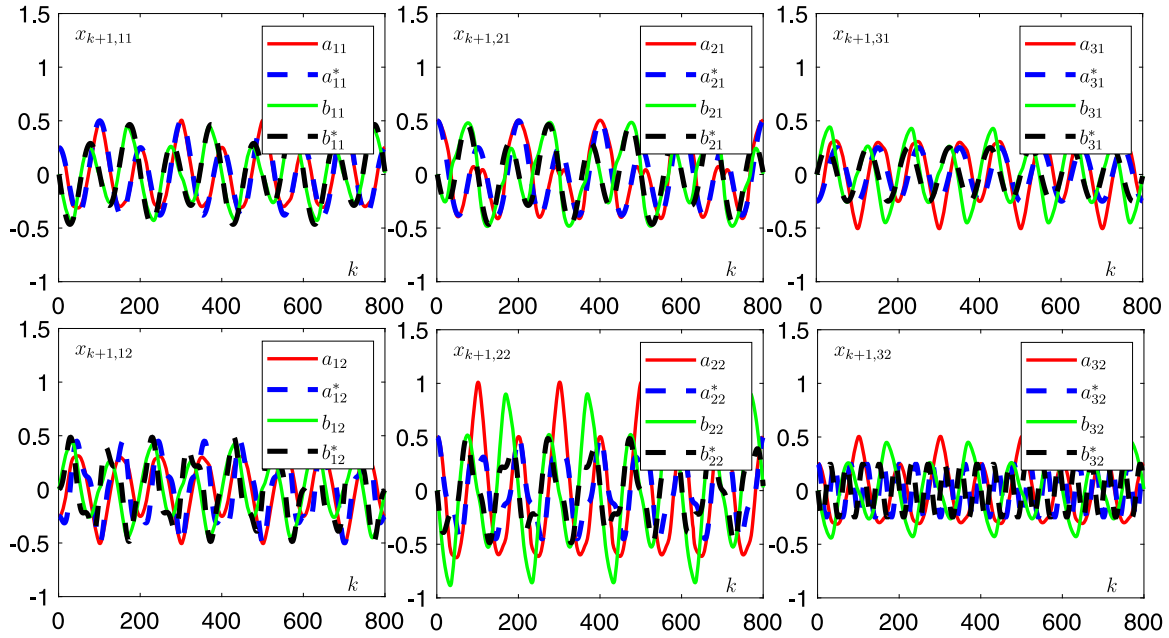


Fig. 4. Element trajectories of theoretical future MPI X_{k+1}^* and state matrix X_{k+1} synthesized by DGMDs algorithm with $\tau = 0.01$ s for computing future MPI problem in Example 2.

respectively. The large MSSREs generated by DGMDs algorithm (31) shown in Table 3 imply that DGMDs algorithm (31) is powerless in Z-type future MPI problem solving.

5. Application to UR10 robot manipulator

In order to verify the feasibility and applicability of O6-DGZND algorithm (12) in the real world, O6-DGZND algorithm (12) is applied to the tracking task on a UR10 redundant manipulator (Yang et al., 2020), and the simulative experimental results are shown in Fig. 5.

We first introduce the UR10 manipulator control. For the UR10 redundant manipulator with six degrees of freedom and working in 3-D space, at time instant $t_k = k\tau \in [0, t_f]$, its end-effector position $\mathbf{r}_{a,k} \in \mathbb{R}^3$ is controlled or manipulated by every joint-angle $\theta_{k,i} \in \mathbb{R}$ with $i = 1, 2, \dots, 6$ via the following functional relationship:

$$\varphi(\theta_k) = \mathbf{r}_{a,k},$$

where $\theta_k = [\theta_{k,1}, \theta_{k,2}, \theta_{k,3}, \theta_{k,4}, \theta_{k,5}, \theta_{k,6}]^T \in \mathbb{R}^6$ denotes the joint-angle vector, and $\varphi(\cdot)$ represents the nonlinear mapping function with known structure and parameters for a given manipulator. Therefore, the relationship between the actual end-effector velocity vector $\dot{\mathbf{r}}_{a,k} \in \mathbb{R}^3$ and joint-velocity vector $\dot{\theta}_k \in \mathbb{R}^6$ can be expressed as $J(\theta_k)\dot{\theta}_k = \dot{\mathbf{r}}_{a,k}$, where Jacobian matrix $J(\theta_k) = \partial\varphi(\theta_k)/\partial\theta_k \in \mathbb{R}^{3 \times 6}$. Given a desired end-effector path $\mathbf{r}_{d,k}$ at time instant t_k , the corresponding joint-velocity vector $\dot{\theta}_k$ can be formulated as $\dot{\theta}_k = J^\dagger(\theta_k)\dot{\mathbf{r}}_{d,k}$, in which the $J^\dagger(\theta_k) \in \mathbb{R}^{6 \times 3}$ is the MPI of the Jacobian matrix $J(\theta_k)$ with full row rank. In order to better achieve tracking results, the following inverse-kinematic problem with error feedback term is considered:

$$\dot{\theta}_k = J^\dagger(\theta_k)(\dot{\mathbf{r}}_{d,k} + \alpha(\mathbf{r}_{d,k} - \varphi(\theta_k))),$$

where $\alpha \in \mathbb{R}^+$ represents the error feedback weight coefficient and we set $\alpha = 1$. Hence, the proposed O6-DGZND algorithm (12) can be adopted to compute the $J^\dagger(\theta_k)$, and further applied to the tracking control of the UR10 robot manipulator.

The tracking task with the following desired path is performed:

$$\mathbf{r}_{d,k} = \begin{bmatrix} 0.08(\cos(2\pi t_k/t_f) + \cos(4\pi t_k/t_f)) + x_0 - 0.16 \\ 0.08(\sin(2\pi t_k/t_f) + \sin(4\pi t_k/t_f)) + y_0 \\ z_0 \end{bmatrix}, \quad (32)$$

where x_0 , y_0 , and z_0 denote the values of the initial position of end-effector along X-axis, Y-axis, and Z-axis, respectively, which is decided

by the initial angle θ_0 . With task duration $t_f = 8$ s, the sampling gap $\tau = 0.001$ s, and the initial angle $\theta_0 = [0, -2\pi/3, -2\pi/3, -\pi/6, 2\pi/3, 0]^T$ rad, the proposed O6-DGZND algorithm (12) is applied on UR10 robot manipulator to conduct the tracking task, and the corresponding simulative experimental results are presented in Fig. 5.

Specifically, Fig. 5(a) shows the motion trajectories synthesized by O6-DGZND algorithm (12), and the resultant trajectories of each joint-angle θ_i with $i = 1, 2, \dots, 6$ are plotted in Fig. 5(c). For better observation, more detailed actual trajectory and desired path are displayed in Fig. 5(b). To compare the difference more intuitively, the trajectory of residual error $e_{k+1} = \|\mathbf{r}_{a,k+1} - \mathbf{r}_{d,k+1}\|_2$ generated by O6-DGZND algorithm (12) is illustrated in Fig. 5(d), where $\|\cdot\|_2$ denotes the 2-norm of a vector. Besides, for comparison, the trajectory of residual error e_{k+1} by O6-DGZND algorithm (12) with $\tau = 0.01$ s is also shown in Fig. 5(d). As shown in Fig. 5, one can conclude that, with the aid of O6-DGZND algorithm (12), the motion tracking task is effectively and precisely performed.

6. Conclusion

Based on the existed ZND and GND model, this paper has proposed the CGZND model without inverse computing to handle the future MPI problem (1). Correspondingly, a new class of DGZND algorithms termed as O2-DGZND through O6-DGZND with different convergence precisions have been developed via combining different ZTD and extrapolation formulas. The 0-stability, consistency, truncation error, and the MSSRE pattern of the five DGZND algorithms have been analyzed by rigorous mathematical deduction. Remarkably, the O6-DGZND algorithm (12) has higher computation precision, and all the proposed DGZND algorithms can be utilized to solve the knotty Z-type future MPI problem effectively while the existed representative DGMDs algorithm (31) cannot. In addition, numerical experiments have been performed to illustrate the efficacy and effectiveness of the proposed DGZND algorithms, and the comparative experimental results have validated the superiority of the O6-DGZND algorithm (12) in computational precision. Lastly, the O6-DGZND algorithm (12) has been successfully employed to a UR10 robot manipulator, which further indicates its practicability. Finally, some potential future research directions are presented as follows. Applying the O6-DGZND algorithm to other kinds of future equation solving problems could be a future research direction.

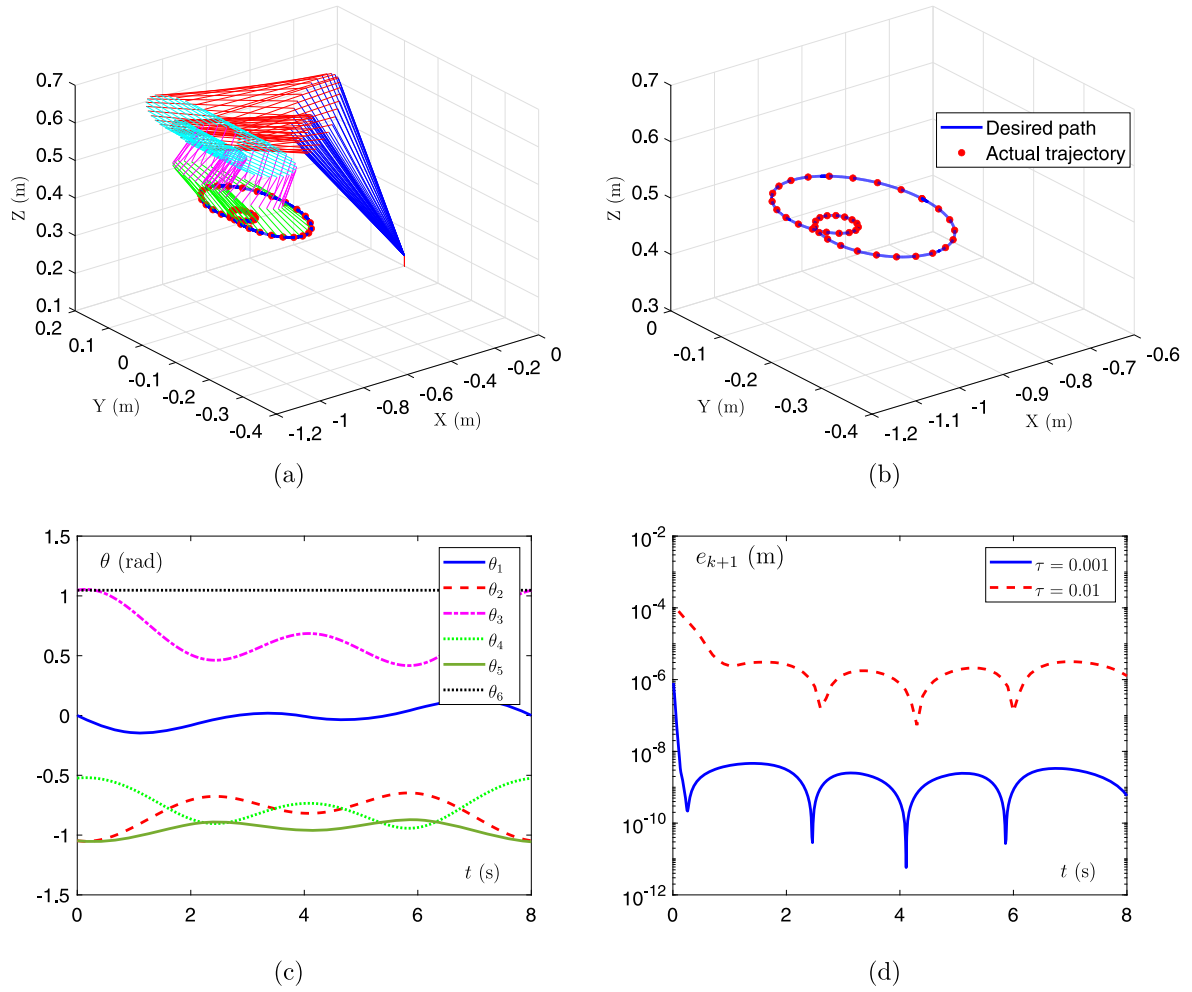


Fig. 5. Simulation results generated by proposed O6-DGZND algorithm (12) for UR10 robot manipulator end-effector tracking desired path (32). (a) Manipulator motion trajectories. (b) Desired path and actual trajectory. (c) Joint-angle trajectories. (d) End-effector tracking error trajectories.

Meanwhile, developing more efficient and accurate time discretization formulas for discretization of continuous models represents a possible future research direction. Furthermore, popularizing the O6-DGZND algorithm in more kinds of engineering applications is also a significant future research direction.

CRediT authorship contribution statement

Wenqi Wu: Methodology, Investigation, Software, Visualization, Writing – original draft. **Yunong Zhang:** Conceptualization, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This work is aided by the National Natural Science Foundation of China (Grant No. 61976230), the Guangdong Province Universities and Colleges Pearl River Scholar Funded Scheme (Grant No. 2018), the

Key-Area Research and Development Program of Guangzhou (Grant No. 202007030004), and the Research Fund Program of Guangdong Key Laboratory of Modern Control Technology (Grant No. 2017B03030-14165).

Appendix A

The proof process of Lemma 3 is provided as below.

Proof. According to the Taylor's expansion of $s(t_{k+1})$, $s(t_{k-1})$, $s(t_{k-2})$, $s(t_{k-3})$, $s(t_{k-4})$, and $s(t_{k-5})$ at time instant t_k , one has

$$s(t_{k+1}) = s(t_k) + \tau \dot{s}(t_k) + \frac{\tau^2}{2!} \ddot{s}(t_k) + \frac{\tau^3}{3!} \dddot{s}(t_k) + \frac{\tau^4}{4!} s^{(4)}(t_k) + \frac{\tau^5}{5!} s^{(5)}(t_k) + O(\tau^6), \quad (33)$$

$$s(t_{k-1}) = s(t_k) - \tau \dot{s}(t_k) + \frac{(-\tau)^2}{2!} \ddot{s}(t_k) + \frac{(-\tau)^3}{3!} \dddot{s}(t_k) + \frac{(-\tau)^4}{4!} s^{(4)}(t_k) + \frac{(-\tau)^5}{5!} s^{(5)}(t_k) + O(\tau^6), \quad (34)$$

$$s(t_{k-2}) = s(t_k) - 2\tau \dot{s}(t_k) + \frac{(-2\tau)^2}{2!} \ddot{s}(t_k) + \frac{(-2\tau)^3}{3!} \dddot{s}(t_k) + \frac{(-2\tau)^4}{4!} s^{(4)}(t_k) + \frac{(-2\tau)^5}{5!} s^{(5)}(t_k) + O(\tau^6), \quad (35)$$

$$s(t_{k-3}) = s(t_k) - 3\tau \dot{s}(t_k) + \frac{(-3\tau)^2}{2!} \ddot{s}(t_k) + \frac{(-3\tau)^3}{3!} \dddot{s}(t_k) + \frac{(-3\tau)^4}{4!} s^{(4)}(t_k) + \frac{(-3\tau)^5}{5!} s^{(5)}(t_k) + O(\tau^6), \quad (36)$$

$$s(t_{k-4}) = s(t_k) - 4\tau \dot{s}(t_k) + \frac{(-4\tau)^2}{2!} \ddot{s}(t_k) + \frac{(-4\tau)^3}{3!} \ddot{\ddot{s}}(t_k) + \frac{(-4\tau)^4}{4!} s^{(4)}(t_k) + \frac{(-4\tau)^5}{5!} s^{(5)}(t_k) + O(\tau^6), \quad (37)$$

and

$$s(t_{k-5}) = s(t_k) - 5\tau \dot{s}(t_k) + \frac{(-5\tau)^2}{2!} \ddot{s}(t_k) + \frac{(-5\tau)^3}{3!} \ddot{\ddot{s}}(t_k) + \frac{(-5\tau)^4}{4!} s^{(4)}(t_k) + \frac{(-5\tau)^5}{5!} s^{(5)}(t_k) + O(\tau^6), \quad (38)$$

where $\dot{s}(t_k)$, $\ddot{s}(t_k)$, $\ddot{\ddot{s}}(t_k)$, $s^{(4)}(t_k)$, and $s^{(5)}(t_k)$ denote the first-, second-, third-, fourth-, and fifth-order time derivatives of $s(t)$ at t_k , respectively. Then, multiplying (33), (34), (35), (36), (37), and (38) with 1, 15, -20, 15, -6, and -1, respectively, yields the O6-E formula (11). Thus, the proof is completed. $\square$

Appendix B

Given a linear N -step method/formula $\sum_{i=0}^N \alpha_i \xi_{k+i} = \tau \sum_{i=0}^N \beta_i \xi_{k+i}$, let $\varphi(\zeta) = \sum_{i=0}^N \alpha_i \zeta^i$ denoting its characteristic polynomial. Then, we have the following three results regarding the linear N -step method/formula are presented (Griffiths & Higham, 2010; Suli & Mayers, 2003).

Result 1. A linear N -step method/formula is 0-stable (i.e., has 0-stability), if all roots ζ of the polynomial characteristic polynomial equation $\varphi(\zeta) = 0$ satisfy the root condition that $|\zeta| \leq 1$ with $|\zeta| = 1$ being simple. In addition, the 0-stability is sometimes called Dahlquist stability or root stability.

Result 2. A linear N -step method/formula is said to be consistent (i.e., has consistency) of order p if the truncation error for the exact solution is of order $O(\tau^{p+1})$ where $p > 0$.

Result 3. A linear N -step method/formula is convergent, i.e., $x_{[t/\tau]} \rightarrow x^*(t)$, for all $t \in [0, t_f]$, as $\tau \rightarrow 0$, if and only if the method is 0-stable and consistent. That is, consistency plus 0-stability leads to convergence, which is also known as Dahlquist equivalence theorem. Besides, a 0-stable consistent method/formula converges with the order of its truncation error.

References

- Chen, J., Guo, J., & Zhang, Y. (2020). General ten-instant DTDMSR model for dynamic matrix square root finding. *Cybernetics and Systems*, 52(1), 127–143.
- Chen, D., Li, S., Lin, F., & Wu, Q. (2020). New super-twisting zeroing neural-dynamics model for tracking control of parallel robots: A finite-time and robust solution. *IEEE Transactions on Cybernetics*, 50(6), 2651–2660.
- Chen, J., & Zhang, Y. (2019). Discrete-time ZND models solving ALMMP via eight-instant general and other formulas of Zead. *IEEE Access*, 7, 125909–125918.
- Chountasis, S., Katsikis, V. N., & Pappas, D. (2009). Applications of the Moore-Penrose inverse in digital image restoration. *Mathematical Problems in Engineering*, 2009, Article 170724.
- Euler, L. (1755). *Institutiones calculi differentialis*. St. Petersburg, Russia: Academiae Scientiarum Petropolitanae.
- Getz, N., & Marsden, J. (1997). Dynamical methods for polar decomposition and inversion of matrices. *Linear Algebra and its Applications*, 258, 311–343.
- Griffiths, D., & Higham, D. (2010). *Numerical methods for ordinary differential equations: initial value problems*. London, UK: Springer-Verlag.
- Guo, D., Nie, Z., & Yan, L. (2017). Novel discrete-time sZhang neural network for time-varying matrix inversion. *IEEE Transactions on Systems, Man, and Cybernetics*, 47(8), 2301–2310.
- Hartwig, R. (1976). Singular value decomposition and the Moore-Penrose inverse of bordered matrices. *Journal of Computational and Applied Mathematics*, 31(1), 31–41.
- Jia, L., Xiao, L., Dai, J., Qi, Z., Zhang, Z., & Zhang, Y. (2020). Design and application of an adaptive fuzzy control strategy to zeroing neural network for solving time-variant QP problem. *IEEE Transactions on Fuzzy Systems*, 29(6), 1544–1555.
- Jin, L., Li, S., Hu, B., Liu, M., & Yu, J. (2019). A noise-suppressing neural algorithm for solving the time-varying system of linear equations: A control-based approach. *IEEE Transactions on Industrial Informatics*, 15(1), 236–246.
- Jin, L., Li, S., Xiao, L., Lu, R., & Liao, B. (2017). Cooperative motion generation in a distributed network of redundant robot manipulators with noises. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 48(10), 1715–1724.

- Jin, L., Wei, L., & Li, S. (2023). Gradient-based differential neural-solution to time-dependent nonlinear optimization. *IEEE Transactions on Automatic Control*, 68(1), 620–627.
- Jin, L., Yan, J., Du, X., Xiao, X., & Fu, D. (2020). RNN for solving time-variant generalized Sylvester equation with applications to robots and acoustic source localization. *IEEE Transactions on Industrial Informatics*, 16(10), 6359–6369.
- Jin, L., & Zhang, Y. (2014). Discrete-time Zhang neural network of $O(\tau^2)$ pattern for time-varying matrix pseudoinversion with application to manipulator motion generation. *Neurocomputing*, 142, 165–173.
- Jin, L., Zheng, X., & Luo, X. (2022). Neural dynamics for distributed collaborative control of manipulators with time delays. *IEEE/CAA Journal of Automatica Sinica*, 9(5), 854–863.
- Kong, Y., Chen, S., Jiang, Y., Wang, H., & Chen, H. (2023). Zeroing neural network with fuzzy parameter for cooperative manner of multiple redundant manipulators. *Expert Systems with Applications*, 212, Article 118735.
- Li, S., & Li, Y. (2014). Nonlinearly activated neural network for solving time-varying complex Sylvester equation. *IEEE Transactions on Cybernetics*, 44(8), 1397–1407.
- Li, J., Mao, M., Uhlig, F., & Zhang, Y. (2019). A 5-instant finite difference formula to find discrete time-varying generalized matrix inverses, matrix inverses, and scalar reciprocals. *Numerical Algorithms*, 81(2), 609–629.
- Li, W., Xiao, L., & Liao, B. (2020). A finite-time convergent and noise-rejection recurrent neural network and its discretization for dynamic nonlinear equations solving. *IEEE Transactions on Cybernetics*, 50(7), 3195–3207.
- Li, Z., & Zhang, Y. (2010). Improved Zhang neural network model and its solution of time-varying generalized linear matrix equations. *Expert Systems with Applications*, 37(10), 7213–7218.
- Liao, B., & Liu, W. (2015). Pseudoinverse-type bi-criteria minimization scheme for redundancy resolution of robot manipulators. *Robotica*, 33(10), 2100–2113.
- Liao, B., & Zhang, Y. (2014). Different complex ZFs leading to different complex ZNN models for time-varying complex generalized inverse matrices. *IEEE Transactions on Neural Networks and Learning Systems*, 25(2), 1621–1631.
- Liu, Y., Liu, K., Wang, G., Sun, Z., & Jin, L. (2023). Noise-tolerant zeroing neurodynamic algorithm for upper limb motion intention-based human-robot interaction control in non-ideal conditions. *Expert Systems with Applications*, 213, Article 118891.
- Liu, M., Zhang, X., Shang, M., & Jin, L. (2022). Gradient-based differential kWTA network with application to competitive coordination of multiple robots. *IEEE/CAA Journal of Automatica Sinica*, 9(8), 1452–1463.
- Petkovic, M., & Stanimirovic, P. (2011). Iterative method for computing the Moore-Penrose inverse based on Penrose equations. *Journal of Computational and Applied Mathematics*, 235(6), 1604–1613.
- Qin, S., Yang, X., Xue, X., & Song, J. (2017). A one-layer recurrent neural network for pseudononconvex optimization problems with equality and inequality constraints. *IEEE Transactions on Cybernetics*, 47(10), 3063–3074.
- Qiu, B., & Zhang, Y. (2019). Two new discrete-time neurodynamic algorithms applied to online future matrix inversion with nonsingular or sometimes-singular coefficient. *IEEE Transactions on Cybernetics*, 49(6), 2032–2045.
- Rao, C., & Mitra, S. (1971). *Generalized inverse of a matrix and its applications*. New York, USA: Wiley.
- Suli, E., & Mayers, D. (2003). *An introduction to numerical analysis*. Cambridge, UK: Cambridge University Press.
- Sun, Z., Wang, G., Jin, L., Cheng, C., Zhang, B., & Yu, J. (2022). Noise-suppressing zeroing neural network for online solving time-varying matrix square roots problems: A control-theoretic approach. *Expert Systems with Applications*, 192, Article 116272.
- Van der Veen, A., Talwar, S., & Paulraj, A. (1997). A subspace approach to blind space-time signal processing for wireless communication systems. *IEEE Transactions on Signal Processing*, 45(1), 173–190.
- Wang, J. (1997). Recurrent neural networks for computing pseudoinverses of rank-deficient matrices. *SIAM Journal on Scientific Computing*, 18(5), 1479–1493.
- Wu, W., & Zhang, Y. (2022). Zeroing neural network with coefficient functions and adjustable parameters for solving time-variant Sylvester equation. *IEEE Transactions on Neural Networks and Learning Systems*, in press, <http://dx.doi.org/10.1109/TNNLS.2022.3212869>.
- Wu, D., Zhang, Y., Guo, J., Li, Z., & Ming, L. (2020). GMDS-ZNN model 3 and its ten-instant discrete algorithm for time-variant matrix inversion compared with other multiple-instant ones. *IEEE Access*, 8, 228188–228198.
- Wu, W., & Zheng, B. (2020). Improved recurrent neural networks for solving Moore-Penrose inverse of real-time full-rank matrix. *Neurocomputing*, 418, 221–231.
- Wu, W., & Zheng, B. (2022). Two new Zhang neural networks for solving time-varying linear equations and inequalities systems. *IEEE Transactions on Neural Networks and Learning Systems*, in press, <http://dx.doi.org/10.1109/TNNLS.2021.3126114>.
- Xiao, L., Cao, Y., Dai, J., Jia, L., & Tan, H. (2021). Finite-time and predefined-time convergence design for zeroing neural network: Theorem, method, and verification. *IEEE Transactions on Industrial Informatics*, 17(7), 4724–4732.
- Xiao, L., He, Y., & Liao, B. (2022). A parameter-changing zeroing neural network for solving linear equations with superior fixed-time convergence. *Expert Systems with Applications*, 208, Article 118086.
- Yang, M., Zhang, Y., Zhang, Z., & Hu, H. (2020). Adaptive discrete ZND models for tracking control of redundant manipulator. *IEEE Transactions on Industrial Informatics*, 16(12), 7360–7368.

- Zhang, Y., Chen, K., & Tan, H. (2009). Performance analysis of gradient neural network exploited for online time-varying matrix inversion. *IEEE Transactions on Automatic Control*, 54(8), 4940–4945.
- Zhang, Y., Jiang, D., & Wang, J. (2002). A recurrent neural network for solving Sylvester equation with time-varying coefficients for solving Sylvester equation with time-varying coefficients. *IEEE Transactions on Neural Networks*, 13(5), 1053–1063.
- Zhang, Z., Kong, L., Zheng, L., Zhang, P., Qu, X., Liao, B., & Yu, Z. (2018). Robustness analysis of a power-type varying-parameter recurrent neural network for solving time-varying QM and QP problems and applications. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 50(12), 5106–5118.
- Zhang, Y., Li, S., Gui, J., & Luo, X. (2018). Velocity-level control with compliance to acceleration-level constraints: A novel scheme for manipulator redundancy resolution. *IEEE Transactions on Industrial Informatics*, 14(3), 921–930.
- Zhang, Y., Li, S., Kadry, S., & Liao, B. (2019). Recurrent neural network for kinematic control of redundant manipulators with periodic input disturbance and physical constraints. *IEEE Transactions on Cybernetics*, 49(12), 4194–4205.
- Zhang, Y., Li, Z., Yang, M., Yu, P., & Tan, N. (2021). Gradient-Zhang neural dynamics models computing pseudoinverses of time-varying matrices via ZeaD and extrapolation formulas. In *2021 International joint conference on neural networks* (pp. 1–9).
- Zhang, Y., Ling, Y., Yang, M., Yang, S., & Zhang, Z. (2022). Inverse-free discrete ZNN models solving for future matrix pseudoinverse via combination of extrapolation and ZeaD formulas. *IEEE Transactions on Neural Networks and Learning Systems*, 32(6), 2663–2675.
- Zhang, Y., Shi, G., Li, J., Wu, G., & Qi, Z. (2018). Zhang matrix found as an exception with its time-dependent pseudoinverse unsolvable by Getz-Masden dynamic system. In *5th International conference on systems and informatics* (pp. 757–762).
- Zhang, W., Wu, J., Yang, Y., & Akilan, T. (2021). Multimodel feature reinforcement framework using Moore-Penrose inverse for big data analysis. *IEEE Transactions on Neural Networks and Learning Systems*, 32(11), 5008–5021.
- Zhang, Y., Yang, Z., Chen, J., Qi, Z., & Yang, G. (2019). Presentation, derivation and numerical experiments of a group of extrapolation formulas. In *9th International conference on information science and technology* (pp. 240–246).
- Zhang, Y., & Yi, C. (2011). *Zhang neural networks and neural-dynamic method*. New York, USA: Nova Science Publishers.
- Zhang, Z., & Zheng, L. (2019). A complex varying-parameter convergent-differential neural-network for solving online time-varying complex Sylvester equation. *IEEE Transactions on Cybernetics*, 49(10), 3627–3639.
- Zhou, J., Zhu, Y., Li, X., & You, Z. (2002). Variants of the Greville formula with applications to exact recursive least squares. *SIAM Journal on Matrix Analysis and Applications*, 24(1), 150–164.